Runhao (Rain) Li

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Education

University of Pennsylvania *Jan 2027 – Expected in Dec 2028*
Master of Science in Robotics Philadelphia, PA

University of Toronto *Sept 2021 – June 2026*
Bachelor of Applied Science in Engineering Science with Honours; Robotics Engineering Toronto, ON

- **cGPA: 3.83/4.00**; Dean's Honour List (all terms); Mathematical & Computational Sciences Research Award.

Research & Industry Experience

Researcher — Robot Learning & Manipulation *Sep 2026 – Present*
LabOne, GRASP Lab, University of Pennsylvania Remote / Philadelphia, PA

- (Ongoing) Investigating action-space criticality by perturbing demonstrated actions to improve learning robustness.

Researcher — Robot Learning & Embodied AI *May 2025 - June 2026*
Toronto Robotics and AI Lab (TRAILab), UTIAS Toronto, ON

- Designed and implemented **AgentVLN**, augmenting a **Vision Language Navigation** backbone with hierarchical **scene-graph memory** for task-conditioned retrieval and persistent spatial understanding; improved R2R-CE success rate from 54.0% to 62.6%.
- As part of *AgentVLN*, developed an **agentic reasoning loop** for landmark tracking, navigation progress monitoring, deviation detection, and recovery; adapted a downstream **reinforcement learning** based locomotion policy to execute VLN decisions as physical Husky motion in Isaac Sim, with reward shaping for obstacle avoidance.
- Contributed to **STaR** , a long-horizon multimodal **robot-memory** framework for **task-conditioned retrieval and embodied reasoning**; refined its 3D memory pipeline and improved cross-frame object association to maintain consistent object identities over time; co-authored the resulting RA-L paper.
- Deployed and evaluated **OpenNav** on a physical Husky robot for indoor/outdoor navigation, integrating multi-sensor perception and resolving synchronization and sensor-fusion issues.

Researcher — Robotics & 3D Vision *May 2025 - Sept 2025*
Robotics Vision & Learning (RVL) Lab, University of Toronto Robotics Institute Toronto, ON

- Proposed and developed a novel **sparse-view 4D Gaussian Splatting reconstruction** method by extending state-of-the-art single-view 4DGS with inter-camera attention and 3D motion-tracklet fusion, reducing required camera views from ∼**20 to** ∼**4** while maintaining the same reconstruction quality on real-world robotic manipulation scenes.
- Contributed to integrating the resulting 4D scene into an **adversarial study** on iterative **Diffusion Policy** learning framework, with controllable **scene augmentation** over objects and appearance to identify policy failure modes and guide targeted retraining on adversarial failure cases.

Co-op Assistant Researcher for Autonomous Driving *May 2024 - May 2025*
Huawei Technologies Canada Markham, ON

- Built and optmized the whole end-to-end implementation of **UniGaussian**, unifying fisheye and pinhole cameras for multi-camera **3D Gaussian Splatting**; co-authored the resulting 3DV 2026 paper.
- Independently proposed and developed a single-image-to-3D pipeline addressing fine-detail degradation in **diffusion-based 3D generation**, using multi-view fusion and optical-flow feature transfer to recover fine-details.
- Enhanced **relightable 3D Gaussian Splatting** by decoupling illumination from scene appearance and geometry, enabling robust reconstruction across varying lighting conditions for autonomous-driving scenes.

Algorithm Developer Intern & System Architect Assistant *May 2023 – Sept 2023*
SenseTime Inc. Shanghai, China

- Developed a **NeRF-based 3D reconstruction** pipeline for crop phenotyping, improving occluded-fruit reconstruction via cross-view consistency and integrating 3D traits into a multimodal fruit-trait prediction pipeline.

Project Highlights

Gesture-Aware Autonomous Human-Following Drone, Undergraduate Capstone *Jan 2026 – Apr 2026*

- Co-built a from-scratch autonomous drone with **Jetson Nano**, RGB/RealSense sensors; **led** developing the **perception to-control** stack including human tracking, closed-loop flight control, and learning-based gesture recognition.

Risk Map Prediction for Autonomous Driving *Jan 2026 – Apr 2026*

- Designed a dense driving risk-map prediction framework using Depth Anything estimates and a custom **depth-gated SegFormer** for RGB Semantic-depth fusion, reaching pixel accuracy 92.3% on unseen datasets.

Skills

- **Programming & ML**: Python, C++, PyTorch, OpenCV, Open3D, MATLAB, C, SystemVerilog
- **Robotics & Tools**: ROS/ROS2, Isaac Sim, Habitat, Gazebo, Genesis, CARLA, Linux, Git, LaTeX